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REVIEW

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Digital twins in cardiology and cardiovascular disease management: a review

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Abstract

Background Cardiovascular diseases remain the leading cause of mortality globally, yet traditional population-based risk stratification and standardised treatment protocols often fail to capture individual patient variability in physiology, disease progression, and therapeutic response. Digital twins, dynamic, patient-specific computational replicas integrating multi-modal clinical data with mechanistic and artificial intelligence models, represent a paradigm shift towards precision cardiovascular medicine.

Aim This narrative review synthesises current evidence on cardiac digital twin technologies, evaluates their clinical applications across cardiovascular subspecialties, and outlines translational pathways from research prototypes to routine clinical implementation.

Methods We conducted a comprehensive narrative review of peer-reviewed literature, regulatory guidance documents, and industry white papers published between 2023 and 2025. Sources included PubMed, Scopus, regulatory agency websites (FDA, EMA), and cardiovascular society position statements. Key search terms encompassed "digital twins", "in-silico cardiology", "virtual heart", "computational modelling", and disease-specific applications.

Results Evidence demonstrates feasibility and emerging clinical utility of digital twins across multiple domains: electrophysiology (atrial fibrillation substrate mapping, ventricular arrhythmia risk stratification), device therapy optimisation (cardiac resynchronisation therapy response prediction), heart failure monitoring (decompensation forecasting), and structural interventions (procedural planning). Regulatory frameworks increasingly accommodate in silico evidence for device development, though prospective randomised controlled trials remain limited.

Conclusion Cardiac digital twins are transitioning from research tools to translational applications with tangible clinical potential. Realising widespread benefit requires prospective outcome validation, interoperable infrastructure, credible uncertainty quantification, and governance frameworks ensuring equitable access and ethical deployment.

Keywords Digital twins, Precision cardiology, Computational modelling, Cardiac electrophysiology, Personalised medicine

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Introduction

The cardiovascular disease burden continues to escalate globally, accounting for approximately 20 million deaths annually and representing nearly one-third of all mortality worldwide [1]. Despite significant advances in pharmacotherapy, interventional techniques, and device-based treatments over recent decades, substantial heterogeneity in treatment response persists across patient populations [2, 3]. Current cardiovascular practice is still dominated by population-derived risk scores and standardised guideline algorithms that provide average rather than truly individualised risk estimates and treatment recommendations, with empirical therapy adjustment often failing to account for patient-specific anatomy, electrophysiology, haemodynamics, and disease substrate [4]. This one-size-fits-all paradigm is reflected in persistent non-response rates of approximately 30–40% after cardiac resynchronisation therapy in real-world cohorts [5], the largely unpredictable individual efficacy and tolerability of antiarrhythmic drugs, and ongoing difficulty in reliably stratifying the risk of heart failure decompensation before overt clinical deterioration, despite the emergence of telemonitoring and AI-based predictive tools [6, 7].

A cardiac digital twin is defined as a patient-specific, dynamically updated computational model of cardiac anatomy, electrophysiology, mechanics, and/or haemodynamics, personalised from multi-modal clinical data and validated against measurable physiological outputs [8, 9]. This distinguishes digital twins from conventional patient-specific simulation, which typically produces a static, single-timepoint model, and from purely data-driven predictive models, which lack explicit mechanistic physiological representation. The defining characteristic of a digital twin is its dynamic personalisation: the model is iteratively updated as the patient's physiology evolves, enabling longitudinal in silico experimentation and virtual therapy testing [10, 11].

Precision medicine promises to address this limitation by tailoring prevention, diagnosis, and treatment strategies to individual patient characteristics [12]. However, cardiovascular medicine presents unique challenges for precision approaches: the heart is a complex electro-mechanical pump operating within a tightly regulated circulatory system, where structural, electrical, and haemodynamic properties interact dynamically over multiple temporal scales from millisecond ion-channel kinetics to years-long remodelling processes [8, 13, 14]. Traditional biomarkers and imaging snapshots provide static representations that may inadequately capture these dynamic, multi-scale interactions. Furthermore, prospective randomised trials required to validate personalised interventions are resource-intensive, time-consuming, and ethically challenging when testing patient-specific strategies that by definition cannot be standardised across cohorts [8, 15].

Digital twin technology offers a potential solution by creating continuously updated, patient-specific computational replicas of the cardiovascular system [16]. These virtual models integrate diverse data streams including cardiac imaging (echocardiography, cardiac magnetic resonance, computed tomography), electrophysiological mapping (electrocardiography, body surface mapping, intracardiac recordings), wearable sensor data, genomic information, and electronic health records with mechanistic biophysical simulations and data-driven artificial intelligence algorithms. By fusing physics-based models of cardiac electrophysiology, mechanics, and fluid dynamics with machine learning approaches for pattern recognition and prediction, digital twins enable in silico experimentation: clinicians can test virtual interventions, predict treatment responses, and explore "what-if" scenarios before committing to real-world therapeutic decisions. This capability extends beyond risk prediction to actionable guidance on device programming, procedural planning, and therapy titration [9, 17].

Recent technological convergences have accelerated digital twin development and clinical translation. Advances in high-resolution cardiac imaging enable detailed anatomical reconstruction with quantification of tissue properties such as fibrosis and myocardial strain. Non-invasive electrocardiographic imaging provides electrical substrate characterisation previously requiring invasive mapping. High-performance computing and cloud infrastructure permit near-real-time solution of complex multi-physics simulations [8, 18]. Artificial intelligence and machine learning enhance model personalisation, parameter inference, and surrogate model development for computational efficiency [19, 20]. Regulatory agencies, particularly the United States Food and Drug Administration and the European Medicines Agency, have published frameworks supporting in silico evidence for medical device evaluation, whilst cardiovascular societies increasingly recognise computational modelling's potential role in clinical decision support [21, 22].

Despite growing enthusiasm and technological capability, cardiac digital twins face significant translational barriers including validation challenges, interoperability constraints, computational demands, ethical considerations around algorithmic bias and data privacy, and uncertainty regarding clinical utility and cost-effectiveness.

Methods

Review design and scope

This narrative review synthesises contemporary evidence on digital twin technologies in cardiovascular medicine, focussing on translational applications, validation frameworks, and implementation pathways. Unlike systematic reviews with exhaustive literature retrieval and meta-analysis, our narrative approach prioritises conceptual synthesis, contextual interpretation, and actionable insights for clinicians and researchers navigating this rapidly evolving field.

Information sources and search strategy

We searched PubMed, Scopus, and Google Scholar databases for peer-reviewed articles published between January 2023 and September 2025, supplemented by pre-2023 seminal works establishing foundational concepts. Search terms included combinations of "digital twin*", "virtual heart", "in silico cardiology", "computational cardiac model*", "patient-specific simulation", "precision cardiology", paired with disease- and intervention-specific terms: "atrial fibrillation", "ventricular arrhythmia", "heart failure", "cardiac resynchronisation therapy", "ablation", "structural heart disease", "coronary flow", and "valve intervention". We also reviewed reference lists of key articles, cardiovascular society statements (American Heart Association, European Society of Cardiology, Heart Rhythm Society), and regulatory guidance documents from the FDA, EMA, and International Medical Device Regulators Forum.

Selection criteria and data extraction

Inclusion criteria:

(1) Original research articles describing cardiac digital twin development, validation, or clinical application;

(2) Reviews and perspectives outlining technological frameworks, clinical use cases, or translational challenges;

(3) Regulatory guidance and position papers addressing in silico evidence, modelling credibility, and digital health integration; and

(4) Industry white papers from established platforms (Living Heart Project, commercial simulation vendors) detailing technical capabilities relevant to clinical translation.

Exclusion criteria: We excluded purely methodological papers without cardiovascular application, animal-only studies without human translation context, and abstracts without full-text availability. Two authors independently screened titles and abstracts, resolving discrepancies through discussion. Article selection was guided by relevance to the defined scope. Full texts were retrieved for all potentially eligible records. Discrepancies in selection were resolved through consensus discussion.

Synthesis and analysis

Given the heterogeneity in digital twin implementations, outcome measures, and reporting standards across studies, formal meta-analysis was not feasible. We employed thematic synthesis, organising evidence by clinical application domain, technological building blocks, validation frameworks, and implementation considerations. We critically appraised evidence quality using established criteria: study design (case reports vs cohort studies vs randomised trials), sample size, comparator selection, outcome relevance, and bias risk. Synthesis emphasised identifying consensus findings, highlighting controversies, and mapping knowledge gaps to guide future research priorities.

Building blocks and enabling technologies

Core components of cardiac digital twins

A comprehensive cardiac digital twin comprises multiple interconnected layers, each contributing essential physiological and clinical information [8]. The anatomical foundation derives from three-dimensional reconstruction of cardiac chambers, great vessels, valves, and coronary arteries using computed tomography (CT), cardiac magnetic resonance imaging (MRI), or three-dimensional echocardiography [18, 23]. Advanced image segmentation algorithms, increasingly augmented by deep learning, extract ventricular geometries, quantify wall thickness, and identify structural abnormalities such as aneurysms, hypertrophy patterns, or congenital defects [18]. Beyond gross anatomy, tissue characterisation through late gadolinium enhancement MRI, T1 mapping, or CT-based attenuation analysis enables identification and quantification of myocardial fibrosis, oedema, and infarct scar critical substrates for arrhythmogenesis and contractile dysfunction [8].

The electrophysiological layer models the heart's electrical activation sequence, from sinoatrial node automaticity through atrial and ventricular depolarisation to repolarisation. Mechanistic models incorporate ion-channel kinetics (sodium, calcium, potassium currents governing action potential morphology), tissue conductivity tensors reflecting myocardial fibre orientation and gap-junction coupling, and heterogeneous properties across transmural layers and anatomical regions [11, 14]. Patient-specific personalisation leverages 12-lead electrocardiograms, body surface potential mapping, or invasive electroanatomical mapping to calibrate conduction velocities, refractory periods, and arrhythmogenic substrate properties. Electrocardiographic imaging (ECGi),

which non-invasively reconstructs epicardial potentials from body surface recordings, has emerged as a powerful tool for personalising electrophysiology twins without requiring invasive procedures [24, 25].

Cardiac mechanics encompasses passive and active myocardial properties, ventricular pressure–volume relationships, and wall stress distributions. Finite element models represent the myocardium as a continuum with anisotropic material properties (fibre, sheet, and cross-fibre stiffness), incorporating active contraction forces driven by calcium transients and length-dependent activation. Personalisation fits model parameters to match patient-specific pressure tracings (from catheterisation or non-invasive surrogates), volumetric changes (from imaging), and strain patterns (from speckle-tracking echocardiography or tagged MRI). Coupling mechanical deformation with electrical activation (electro-mechanical coupling) enables simulation of phenomena such as mechanical dispersion influencing cardiac resynchronisation therapy outcomes [20, 25].

Haemodynamics and fluid dynamics extend the digital twin to blood flow within cardiac chambers, across valves, and through coronary vessels. Computational fluid dynamics (CFD) models solve Navier–Stokes equations to simulate flow patterns, shear stresses, and pressure gradients, whilst reduced-order lumped-parameter models (representing the circulation as interconnected resistances, compliances, and inertances) enable efficient simulation of systemic and pulmonary vascular beds [25]. Coronary flow models, particularly relevant for virtual fractional flow reserve (vFFR) applications, simulate pressure drops across stenoses and collateral circulations. Valve models incorporate leaflet kinematics, regurgitant jets, and stenotic gradients, supporting pre-procedural planning for transcatheter interventions [27, 28].

Finally, the integration layer fuses these components with longitudinal clinical data, laboratory values (natriuretic peptides, troponin, renal function), medications, comorbidities, genomic variants influencing drug metabolism or disease susceptibility, and continuous streams from wearable devices (heart rate variability, activity levels, weight) [20]. Machine learning algorithms identify patterns predicting clinical events (heart failure hospitalisation, arrhythmia occurrence) and update model parameters as physiology evolves, transforming static simulations into dynamic, learning systems. The cardiac digital twin integrates these layers through a structured workflow (Fig. 1).

Multi-modal data fusion and interoperability

Constructing a high-fidelity digital twin requires seamless integration of heterogeneous data sources. Typical workflows begin with automated anatomical reconstruction from cardiac CT or MRI via deep learning segmentation, producing geometric scaffolds for simulation [19, 29]. Electrophysiological data integration spans multiple scales: 12-lead ECG informs baseline conduction parameters; body surface potential mapping or ECGi reconstructs epicardial electrical activity; and invasive electroanatomical mapping, where available, provides millimetre-scale voltage and activation data [30, 31]. Mechanical properties are inferred from

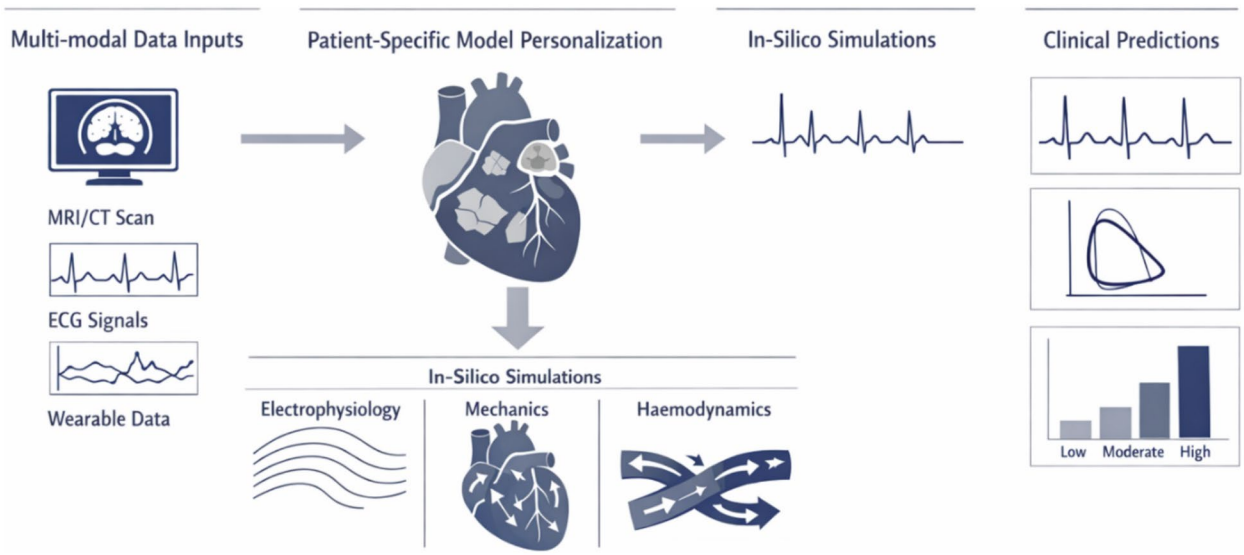


Fig. 1 Schematic workflow of cardiac digital twin construction, integrating imaging, electrophysiological, and physiological data to generate patient-specific simulations for precision cardiology applications across electrophysiology, heart failure, and device optimisation

Table 1 Key enabling technologies and platforms for cardiac digital twins

Platform/technology	Developer/organisation	Core capabilities	Clinical translation status
Living heart project [8]	Dassault Systèmes, FDA, academia	Multi-physics whole-heart model; electro-mechanical coupling; device interaction simulation; AI-powered parametric personalisation	Device R&D validation; emerging patient-specific applications
OpenCARP [19, 39]	International collaboration	Open-source cardiac electrophysiology solver; cellular to organ-scale simulations; GPU acceleration	Research tool; AF and VT substrate mapping studies; increasing clinical research adoption
cHeart/FEniCS [18]	University College London, others	Finite element cardiac mechanics; fluid–structure interaction; patient-specific geometry pipelines	Primarily research; valve intervention planning demonstrations
Electrocardiographic imaging (ECGi) [14]	Multiple vendors (CardioInsight, EP Solutions)	Non-invasive epicardial potential mapping from 250 + body surface electrodes	Clinical use for AF/VT substrate characterisation; FDA-cleared devices available; twin personalisation data source
CT-derived fractional flow reserve (CT-FFR) [18]	HeartFlow, Siemens, others	Coronary flow simulation from coronary CT angiography; stenosis haemodynamic significance	Widespread clinical adoption; NICE-approved in UK; cost-effective alternative to invasive FFR
Cloud high-performance computing [17]	AWS, Google Cloud, Azure	Scalable compute for multi-physics simulations; containerised solvers; near-real-time inference with GPU acceleration	Enabling infrastructure; regulatory considerations for data security and provenance
FHIR/HL7 interoperability standards [20]	Health Level Seven International	Standardised health data exchange; SMART-on-FHIR apps for EHR integration	Foundation for clinical workflow integration; adoption increasing but incomplete

tagged MRI strain imaging, speckle-tracking echocardiography, or invasive pressure–volume loops [32]. Wearable sensor streams, electronic health records, laboratory values, medications, and genomic data contribute longitudinal inputs, increasingly extracted via natural language processing [20, 33].

Achieving seamless data flow demands interoperability standards. Fast Healthcare Interoperability Resources (FHIR), developed by Health Level Seven International, defines standardised APIs for exchanging healthcare data, increasingly adopted in electronic health record systems. SMART-on-FHIR enables third-party applications, potentially including digital twin platforms, to access EHR data with patient consent [34–36]. DICOM standards govern medical image storage, though extending DICOM to encompass simulation outputs and twin-specific metadata remains an area of active development [37, 38]. Despite progress, barriers persist: proprietary simulation platforms use incompatible file formats; imaging systems from different vendors encode metadata inconsistently; and wearable devices lack standardised APIs for clinical-grade data export.

Simulation platforms and computational infrastructure

Table 1 summarises key enabling technologies and platforms advancing cardiac digital twin development and deployment.

The Living Heart Project exemplifies industrial-scale digital twin development, providing validated multi-physics models used by medical device manufacturers for virtual testing of pacemakers, defibrillators, valves, and structural heart devices. Its 2025 evolution towards AI-augmented parametric twins signals a shift from single-patient models requiring extensive manual personalisation towards scalable platforms enabling rapid virtual cohort generation for device optimisation and regulatory submissions [8, 18].

Computational demands remain substantial: high-fidelity electrophysiology simulations resolving cellular action potential dynamics across millions of myocardial elements require hours to days on traditional workstations [17, 19]. Recent advances in GPU acceleration, reduced-order modelling, and AI-based surrogate models (physics-informed neural networks replacing CFD calculations, for example) aim to compress simulation times to minutes, essential for intra-procedural guidance or emergency decision support. Cloud infrastructure enables on-demand scaling whilst raising questions about data security, latency, and regulatory compliance for medical-grade computations [19, 20].

Artificial intelligence and machine learning integration

Hybrid digital twins combining mechanistic biophysical models with data-driven AI offer synergistic advantages. Physics-based models ensure physiological plausibility and enable causal reasoning ("what causes this arrhythmia substrate?") but require extensive personalisation and computational resources [8, 19, 20]. Machine learning excels at pattern recognition, rapid inference, and learning from large datasets but may lack interpretability and extrapolate poorly beyond training distributions. Integration strategies include:

- Surrogate modelling: Training neural networks to approximate expensive simulations (e.g. a CFD-trained network predicting coronary pressure drops from geometry features), enabling real-time "what-if" analysis during procedures [18, 19].
- Parameter inference: Using inverse methods (e.g. physics-informed neural networks, Bayesian optimisation) to estimate patient-specific model parameters (conduction velocities, stiffness) from clinical observations (ECG, echo strain), reducing manual calibration [18, 19].
- Outcome prediction: Combining mechanistic simulation outputs (virtual pacing activation sequences) with clinical features in ML models to predict treatment response (CRT responders vs non-responders), potentially outperforming either approach alone [14, 17].
- Data assimilation: Sequential learning algorithms (Kalman filters, particle filters) continuously updating twin parameters as new data arrive, maintaining fidelity as disease progresses or therapies change [8, 20].

Generative AI, particularly large language models and multi-modal foundation models, represents an emerging frontier. Potential applications include automated clinical note extraction for twin parameterisation, natural language interfaces for clinician interaction with simulations, and synthetic data generation for training robust models whilst preserving patient privacy [20].

Clinical applications

The following sections critically appraise what has been demonstrated in published studies, not merely what digital twins could theoretically achieve. For each domain, we report available information on study design, sample size, validation approach, reference standards, and, critically, whether patient-centred clinical outcomes have been evaluated, not only technical modelling performance.

Electrophysiology: atrial fibrillation

Atrial fibrillation (AF), the most common sustained cardiac arrhythmia, affects over 50 million individuals globally and confers substantial morbidity through stroke, heart failure, and reduced quality of life [40, 41]. Catheter ablation targeting pulmonary vein isolation and, increasingly, atrial substrate modification represents a cornerstone treatment, yet single-procedure success rates remain approximately 50–70% for persistent AF, with significant inter-patient variability driven by differing fibrosis distributions, atrial geometry, and electrophysiological properties [14, 17]. Digital twins of the left atrium synthesise patient-specific anatomy from cardiac imaging, incorporate fibrosis extent and distribution from late gadolinium enhancement MRI or voltage mapping, and embed fibre orientation and conduction anisotropy inferred from imaging or ECGi data [17, 18].

Several concrete studies illustrate the current state of evidence. Sakata et al published a multicentre study applying left atrial digital twins to stratify persistent AF patients for ablation, demonstrating that twin-guided lesion selection significantly reduced unnecessary ablation of non-arrhythmogenic tissue compared to conventional approaches; however, the study primarily reported computational endpoints (ablation target prediction) with limited long-term patient follow-up data on arrhythmia recurrence or quality-of-life outcomes [42]. A comprehensive scoping review by Karakasis et al synthesised AF twin applications across stroke risk refinement, antiarrhythmic drug response prediction, and ablation strategy optimisation, but noted that most contributing studies were proof-of-concept feasibility analyses in cohorts of fewer than 50 patients, predominantly single-centre and retrospective in design [43].

A critical distinction is warranted between technical modelling performance and patient-centred clinical benefit. Whilst digital twins have demonstrated the ability to recapitulate atrial activation sequences and identify computationally predicted arrhythmogenic drivers, prospective randomised comparisons evaluating whether twin-guided ablation improves freedom from AF recurrence, reduces procedural complications, shortens procedure time, or improves quality of life, compared to conventional electroanatomical mapping, are absent from the current literature. The clinically meaningful endpoint for AF ablation is AF-free survival at 12 months post-procedure; to date, no published randomised controlled trial has used this as a primary outcome in twin-guided versus standard ablation [42, 44]. Computational predictions must be considered hypothesis-generating rather than practice-changing until such trials are completed. Challenges include computational cost (hours per simulation), limited prospective validation, and debate regarding optimal fibrosis quantification methods [43, 45].

Electrophysiology: ventricular arrhythmias

Ventricular tachycardia (VT) in structural heart disease, post-infarction scar, arrhythmogenic cardiomyopathy, non-ischaemic cardiomyopathy, poses similar substrate heterogeneity [14, 45]. Patient-specific ventricular twins reconstruct scar geometry, border zones, and surviving channels from MRI, simulate reentrant circuits, and identify critical isthmus sites for ablation.

The most concrete published evidence comes from two landmark studies. Zhang et al applied genotype-specific heart digital twins to 11 patients with arrhythmogenic right ventricular cardiomyopathy (ARVC), demonstrating that patient-specific computational models incorporating genotype-dependent tissue properties could predict VT circuit locations with spatial concordance to invasive electroanatomical mapping in 9 of 11 cases [46]. This single-centre feasibility study used invasive mapping as the reference standard; clinical outcomes (VT recurrence post-ablation, complications) were not the primary endpoint. Waight et al extended this approach to scar-dependent VT from multiple aetiologies, demonstrating that digital twin-predicted reentrant circuit features matched invasive mapping findings and predicted ablation lesion locations; again, this was a retrospective validation study focussed on anatomical concordance rather than prospective patient outcomes [47].

Critically, no randomised controlled trial has compared digital twin-guided VT ablation against conventional substrate mapping strategies using clinical endpoints such as VT recurrence, hospitalisation, or survival. Published evidence is limited to single-centre feasibility and retrospective validation studies. The gap between demonstrating that a digital twin can identify an arrhythmia circuit and demonstrating that twin-guided ablation improves long-term patient outcomes remains significant and must be addressed in future prospective trials.

Heart failure management: monitoring, prediction, and therapy titration

Heart failure (HF) management demands continuous risk assessment and timely intervention to prevent decompensation. Emerging cardiovascular digital twin frameworks offer a proactive alternative by constructing patient-specific virtual models of cardiac and vascular function that are iteratively updated using multi-modal data, extending existing multiparametric HF remote-monitoring paradigms to detect early deviations from individual baseline physiology and support pre-emptive therapy adjustment [8, 20, 48, 49].

Gu et al published a notable study applying digital twin identification methods to HF patients, demonstrating that patient-specific model parameters could be extracted from routine clinical data and used to stratify patients by haemodynamic profile, informing prognosis; however, the study was primarily methodological and did not evaluate twin-guided therapy titration against standard care in a controlled setting [50]. Geddes et al developed digital twins for non-invasive measurement of predictive markers of right heart failure, demonstrating in a retrospective validation cohort that computationally derived right heart haemodynamic parameters correlated with invasive catheterisation measurements, a technically important validation study, though patient numbers and independent validation cohorts were not large enough for definitive clinical conclusions [51]. Earlier multiparameter remote-monitoring algorithms, though not mechanistic twins, have achieved sensitivities and specificities typically in the 70–85% range for predicting HF hospitalisation within 30 days; however, high false-positive rates and alarm fatigue continue to limit clinical adoption [8, 48, 51, 52].

Regarding patient-centred outcomes: to date, no published randomised trial has evaluated whether digital twin-guided therapy adjustment, compared to guideline-directed medical therapy, reduces HF hospitalisation rates, emergency department visits, or mortality. The clinically meaningful question is not only 'can a digital twin predict decompensation?' but 'does acting on twin-derived alerts change patient trajectories?' This distinction between predictive performance and actionable clinical benefit remains a fundamental evidence gap in the HF digital twin literature [20, 50, 53].

Device therapy optimisation: cardiac resynchronisation therapy

Cardiac resynchronisation therapy (CRT), delivering biventricular pacing to correct electrical dyssynchrony in heart failure patients with wide QRS complexes, improves symptoms, functional capacity, and survival in responders, yet approximately 30–40% of guideline-eligible patients fail to respond [5, 54]. Response heterogeneity reflects variability in electrical substrate, mechanical substrates, lead positioning, and pacing vector configurations.

CRT optimisation represents one of the most clinically advanced digital twin applications, with the highest-quality prospective evidence in the field. Koopsen et al, conducted a prospective single-centre study in 45 CRT candidates, performing pre-implant digital twin simulations predicting response based on virtual left ventricular lead placements [55]. The twin integrated cardiac MRI-derived geometry, fibrosis characterisation, ECG-based electrophysiology personalisation, and finite element mechanical modelling. Digital twin predictions achieved strong correlation for response classification, defined as left ventricular end-systolic volume reduction at 6 months post-CRT, a clinically meaningful, guideline-endorsed endpoint [55]. Importantly, in patients receiving leads at digitally predicted optimal sites, response rates approached 85%, compared to approximately 55–65% in non-twin-guided historical cohorts at the same centre, suggesting procedural planning impact on patient outcomes [5, 14, 17]. The Leclercq et al MORE-CRT MPP trial (Europace 2023) provided additional context on CRT non-responder conversion, informing baseline rates against which digital twin approaches are benchmarked [5].

The CRT evidence from Koopsen et al represents the clearest example to date of a digital twin application demonstrating prospective prediction of a patient-centred outcome (volumetric reverse remodelling, a validated surrogate of clinical response). Limitations include single-centre design, modest sample size ($n=45$), and absence of randomised comparison against standard non-twin-guided implantation. Larger multicentre randomised controlled trials are needed to confirm whether twin-guided lead placement reduces non-responder rates in diverse patient populations and translates to improvements in symptoms, exercise capacity, and survival [55].

Coronary disease and functional assessment

Although not always termed "digital twins", CT-derived fractional flow reserve (CT-FFR) technologies exemplify successful clinical translation of patient-specific computational modelling. CT-FFR applies computational fluid dynamics to coronary CT angiography images, simulating hyperaemic flow and pressure fields to non-invasively estimate FFR values [18, 56].

CT-FFR has the strongest clinical evidence of any cardiovascular digital twin-adjacent technology. The ADVANCE registry (prospective, multicentre, $n>5000$ patients across 38 sites) demonstrated that CT-FFR-guided management safely reduced unnecessary coronary angiography, with outcomes non-inferior to invasive FFR-guided care [56, 57]. The NXT trial (prospective, $n=254$) established diagnostic accuracy exceeding 80% against invasive FFR as reference standard. The UK FORECAST trial (RCT, $n=1400$) confirmed cost-effectiveness and reduction in downstream testing in stable chest pain. The UK National Institute for Health and Care Excellence (NICE) endorsed CT-FFR for stable chest pain evaluation, and reimbursement pathways exist in multiple healthcare systems [18, 56]. This evidence base, spanning prospective registries, RCTs, and health economic analyses, establishes that patient-specific computational

models can reach clinical adoption, improve patient pathways (avoiding unnecessary invasive procedures), and demonstrate cost-effectiveness. It sets the benchmark for other digital twin applications.

Extending beyond stenosis assessment, emerging coronary digital twins incorporate plaque composition, wall shear stress distributions predicting plaque progression sites, and longitudinal remodelling to forecast myocardial infarction risk. Coupling coronary flow models with myocardial perfusion and mechanics enables integrated ischaemia assessment, potentially superseding separate anatomical and functional imaging tests [1, 26, 58].

Structural heart interventions and surgical planning

Transcatheter aortic valve replacement (TAVR), mitral valve repair/replacement, left atrial appendage occlusion, and septal defect closure procedures involve navigating complex patient-specific anatomy with millimetre-scale precision to ensure device apposition, minimise paravalvular leak, and avoid complications [18]. Digital twins reconstructing valve annuli, leaflets, calcium distribution, and adjacent structures enable virtual device deployment [18]. Finite element simulations model device expansion, interaction with native tissue, and resultant haemodynamics.

Published evidence in this domain remains predominantly at the device development and feasibility stage. Skalidis et al comprehensively reviewed digital twins and simulations in transcatheter coronary and structural heart interventions, concluding that virtual device deployment shows promise for predicting paravalvular leak and sizing in complex anatomies, but that prospective clinical studies evaluating patient outcomes, procedure success rates, complication incidence, 30-day mortality, using twin-guided versus standard CT-based planning are lacking [18]. The Living Heart Project's device-interaction libraries have generated extensive regulatory-grade simulation data for device R&D, enabling virtual testing under varied anatomical configurations; however, translation to routine patient-specific procedural planning remains limited to complex or high-risk cases at specialist centres.

The clinically meaningful questions for structural intervention twins are whether pre-procedural digital twin planning reduces rates of paravalvular regurgitation, conduction block, need for permanent pacemaker implantation, or 30-day major adverse cardiac events compared to conventional CT-based planning. These patient-centred outcomes remain largely unevaluated in prospective comparative studies. Table 2 summarises the clinical evidence landscape across all digital twin application domains.

These applications span electrophysiology, heart failure, device therapy, coronary assessment, and structural interventions (Fig. 2).

Evidence and validation framework

Hierarchy of evidence and current status

The evidence base for cardiac digital twins spans a continuum from in silico feasibility demonstrations to prospective clinical trials, with most applications currently residing in early translational stages [24, 59]. Feasibility studies establish technical capability: patient-specific models can be constructed from clinical data within acceptable timeframes; simulations recapitulate observed physiology; and computational predictions correlate with clinical measurements. Numerous publications confirm these proof-of-principle milestones across AF, VT, CRT, and HF domains [20, 24].

Accuracy and validation studies compare digital twin predictions against independent clinical observations. Reported concordance varies by application and metric: electrical activation maps typically achieve 70–90% spatial agreement with clinical mapping; virtual FFR demonstrates > 80% diagnostic accuracy; CRT response predictions reach 80–85% sensitivity and specificity. Importantly, these validation exercises often use retrospective data, introducing potential overfitting and optimism bias [14, 17, 60].

Prospective clinical utility trials remain scarce. The Koopsen et al. CRT cohort ($n = 45$) represents a notable exception, demonstrating prospective prediction accuracy [55]. CT-FFR technologies underwent extensive prospective validation across multiple RCTs, enabling regulatory approval and adoption, a model for other digital twin applications. Ongoing trials include CRT lead placement optimisation studies and AF ablation strategy comparisons [17, 18, 42].

Regulatory pathways and in silico evidence

Medical device regulation traditionally demands human clinical trial evidence demonstrating safety and effectiveness. In silico clinical trials using virtual patient cohorts derived from digital twin platforms offer a complementary evidence source. Regulatory acceptance hinges on establishing model credibility: confidence that simulations adequately represent real-world physiology and device interactions [8, 22].

High-risk decisions demand rigorous verification (solving equations correctly), validation (matching real-world data), and uncertainty quantification. Credibility elements include verification (mathematical correctness), validation (comparing simulations to experimental/clinical data), uncertainty quantification (characterising input

Table 2 Clinical evidence landscape for cardiac digital twin applications

Application domain	Representative studies	Evidence quality	Key findings	Patient-centred outcomes evaluated?	Translational readiness
Electrophysiology (AF) [42, 43]	Sakata 2025, multicentre; Karakasis 2025 review of < 50-patient cohorts	Proof-of-concept cohorts; feasibility studies	Twins recapitulate atrial activation; simulated ablations identify arrhythmogenic sites	No: AF-free survival, QoL not reported in RCTs	Moderate; prospective RCTs absent
Electrophysiology (VT) [46, 47]	Zhang 2023 eLife (ARVC, $n = 11$); Waight 2025 (scar-dependent VT, retrospective)	Case series; single-centre retrospective validation	Patient-specific twins identify reentrant circuits; spatial concordance with invasive mapping	No: VT recurrence, survival not evaluated	Low-to-moderate; outcome trials absent
Heart Failure [50, 51]	Gu 2025; Geddes 2025 (right HF markers); retrospective validation cohorts	Retrospective validation; small prospective pilots	70–85% sensitivity/specificity for 30-day HF hospitalisation prediction; virtual titration feasible	No: RCTs comparing twin-guided vs standard HF care absent	Moderate; prospective intervention trials needed
CRT optimisation [55]	Koopsen 2023, EP Europace ($n = 45$, prospective single-centre); Leclercq MORE-CRT MPP	Prospective cohort; prediction accuracy 80–85%	Digital twin-predicted optimal sites correlate with ≥ 15% LVESV reduction at 6 months	Partial: LVESV reduction (validated surrogate) measured; no RCT vs standard implantation	Moderate-to-high; larger RCTs planned/underway
Coronary disease (CT-FFR) [18, 56]	ADVANCE ($n > 5{,}000$, multicentre); NXT ($n = 254$); FORECAST ($n = 1400$ RCT)	Multi-centre RCTs; real-world registries	Non-inferior to invasive FFR; reduced downstream testing; cost-effective	Yes: catheterisation rates, major adverse cardiac events, cost-effectiveness	High; clinical adoption; regulatory approval (FDA, NICE)
Structural heart [18]	Skalidis 2025; Living Heart Project device libraries	Device development simulations; small patient-specific cohorts	Accurate device–anatomy interaction prediction; clinical utility for complex cases	No: paravalvular regurgitation, complications, mortality not evaluated prospectively	Moderate; primarily device R&D; patient-specific planning emerging

ARVC arrhythmogenic right ventricular cardiomyopathy, *CRT* cardiac resynchronisation therapy, *CT-FFR* CT-derived fractional flow reserve, *HF* heart failure, *LVESV* left ventricular end-systolic volume, *QoL* quality of life, *RCT* randomised controlled trial, *VT* ventricular tachycardia

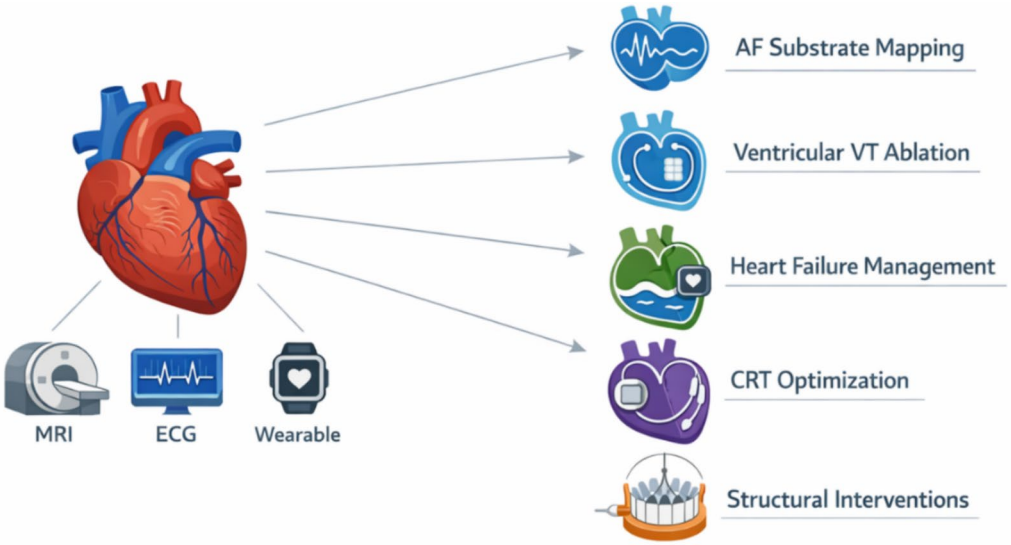


Fig. 2 Overview of clinical applications of cardiac digital twins in key cardiovascular domains, including electrophysiology (atrial fibrillation and ventricular tachycardia), heart failure monitoring, cardiac resynchronisation therapy optimisation, and structural heart interventions

parameter uncertainty), and applicability (ensuring the model's intended use matches validation evidence) [22].

The FDA's Medical Device Development Tools programme created a pathway for qualifying computational models, though no cardiac digital twin platforms have yet achieved formal qualification. Meanwhile, in device R&D, the Living Heart Project's use in pre-clinical virtual testing has become commonplace, reducing physical prototype iterations and animal testing [20, 22]. The EU Medical Device Regulation and In Silico World consortium have outlined parallel frameworks emphasising model transparency, clinical validation, and post-market surveillance [22, 61].

Credibility challenges and best practices

Despite frameworks, cardiac digital twin validation faces substantial challenges. Patient-specific models contain numerous free parameters that may be underconstrained by available clinical measurements, leading to multiple parameter sets producing similar outputs-a non-uniqueness problem. Bayesian approaches quantifying parameter uncertainty and sensitivity analyses identifying influential parameters mitigate this [19, 20, 24]. Cardiovascular physiology evolves over days, months, and years; validating predictive accuracy requires longitudinal follow-up that most published studies lack [20, 50].

Arrhythmia occurrence, sudden cardiac death, and heart failure hospitalisation are relatively rare events. Validating digital twin predictions of these outcomes demands large cohorts and extended follow-up. Furthermore, twins personalised predominantly on white, male, Western populations may perform poorly in under-represented demographic groups, a critical equity consideration [8, 43].

Best-practice recommendations include: pre-specifying validation protocols before model development; employing independent test cohorts distinct from development datasets; reporting uncertainty bounds alongside point predictions; conducting sensitivity analyses across physiological parameter ranges; and engaging clinical end-users throughout development [8, 22, 43].

Regulatory landscape and in silico trials

Current regulatory position

Regulatory agencies globally are evolving their stances on computational modelling as medical evidence. The FDA's Center for Devices and Radiological Health (CDRH) has published multiple guidance documents culminating in the comprehensive Assessing the Credibility of Computational Modelling and Simulation framework, which articulates a risk-based approach: higher-risk decisions necessitate more rigorous credibility evidence than lower-risk supportive uses [62].

The FDA's Medical Device Development Tools (MDDT) programme provides a pathway for formally qualifying computational models [63]. Whilst primarily applied to biomechanical test methods, the framework is being explored for digital twin platforms. The European Medicines Agency (EMA) and Medicines and Healthcare products Regulatory Agency (MHRA) have published complementary guidance. The In Silico World initiative

advocates for frameworks enabling in silico clinical trials as adjuncts or, in limited scenarios, alternatives to traditional trials [64, 65].

In silico clinical trials: promise and practicalities

In silico clinical trials use computational models to generate virtual patient populations, simulate interventions, and predict outcomes. For cardiac digital twins, applications span device optimisation (virtual cohorts testing CRT lead configurations, TAVR valve sizes), drug response prediction, and trial design optimisation [33, 66].

Realising this vision requires virtual cohorts that capture anatomical, physiological, and demographic variability reflecting target patient populations, and rigorous validation that synthetic cohorts reproduce observed outcome distributions [60]. Hybrid trial designs represent a pragmatic intermediate step: using digital twins to enrich trial populations, simulate control arms, or provide mechanistic insights explaining observed clinical effects [67]. Near-term reality likely involves in silico evidence as one component of comprehensive regulatory submissions, complementing rather than replacing clinical investigation.

Post-market surveillance and continuous learning

Digital twin platforms deployed in clinical practice generate real-world data continuously: predictions are logged, clinical outcomes observed, and discrepancies analysed. Machine learning algorithms can ingest this feedback, updating model parameters to enhance accuracy, creating continuously learning systems [68, 69]. Regulatory frameworks are beginning to accommodate adaptive algorithms through pre-specified change control plans. Ensuring updates do not inadvertently introduce biases or degrade performance for subgroups requiring robust governance [70].

Ethical considerations and equity

Algorithmic bias and health disparities

Cardiac digital twins trained or personalised on datasets skewed towards specific demographic groups risk systematically reinforcing existing cardiovascular health inequities [71, 72]. Historical under-representation of women, ethnic minorities, and lower socioeconomic populations in cardiovascular research means imaging libraries, electrophysiology parameter distributions, and outcomes databases disproportionately reflect white, male, Western populations [71, 73]. Digital twins calibrated to these distributions may exhibit lower prediction accuracy and yield suboptimal therapy recommendations for under-represented groups.

Addressing algorithmic bias requires: ensuring training datasets encompass representative demographics; validating twins across subgroups and reporting stratified performance metrics; incorporating social determinants and genomic diversity explicitly in models; and engaging communities affected by health disparities in technology development and governance [73–75].

Transparency, interpretability, and clinician trust

Clinicians integrating digital twin recommendations into practice require trust, confidence that predictions are accurate, and understanding of how conclusions are derived. Hybrid mechanistic–AI twins offer partial mitigation: physics-based components provide interpretable, causal insights [18, 50]. Best practices for transparency include explainability tools (saliency maps, counterfactual simulations), uncertainty communication (displaying prediction confidence intervals), and audit trails (logging data inputs, model versions, and predictions) [31, 73].

Privacy, security, and data governance

Digital twins require sensitive patient data: high-resolution anatomical imaging, genetic information, continuous wearable streams, and comprehensive health records. Privacy concerns span re-identification risks, data breach vulnerabilities, and secondary use consent questions [20, 74]. Technical safeguards include encryption, access controls, anonymisation, and secure multi-party computation. Regulatory compliance encompasses GDPR, HIPAA, and emerging AI-specific regulations [45, 74].

Access and digital divide

High-fidelity digital twins demand advanced imaging, specialised computational infrastructure, and expert multidisciplinary teams, resources concentrated in tertiary academic centres [20, 74]. Mitigation strategies include: technology democratisation via open-source platforms and cloud-based software-as-a-service models; hub-spoke telemedicine models where community sites transmit data to centralised computational hubs; value-based reimbursement demonstrating cost-reduction; and global partnerships sharing resources across institutions and regions [31, 73, 74].

Implementation roadmap

Institutional readiness assessment

Centres planning cardiac digital twin implementation must perform thorough readiness assessments across clinical, technical, and operational domains. Clinical need identification should pinpoint patient cohorts where digital twins provide clear, demonstrated value, such as high CRT non-responder rates, complex AF substrates, or elevated HF rehospitalisation risks [20, 31, 50].

Infrastructure assessment must evaluate imaging capabilities (cardiac MRI with late gadolinium enhancement, ECGi systems, coronary CT), computational resources (on-premise HPC or cloud access, GPU workstations), data integration architecture (PACS connectivity, EHR APIs), and personnel expertise. Gaps may require capital investments, training programmes, or external partnerships [74, 76].

Starting with focussed use cases

Rather than attempting comprehensive cardiovascular digital twin platforms ab initio, pragmatic implementation begins with targeted, evidence-supported applications. CRT virtual pacing planning represents a high-priority starting point, driven by prospective cohort evidence demonstrating prediction accuracy exceeding 80%, feasible pre-implant workflow integration, and a clinically meaningful primary endpoint (LVESV reduction) [20, 31, 55]. AF ablation target refinement in persistent AF or redo procedures follows, leveraging left atrial geometry and fibrosis characterisation increasingly routine in AF workup [20, 50, 76]. Table 3 outlines pragmatic implementation priorities.

Building data and computational infrastructure

Robust digital twin implementation requires standardised, reproducible data pipelines. Institutions should develop data dictionaries and protocols (standardising imaging sequences for reproducibility), automated processing pipelines (minimising manual steps via AI-based segmentation and containerised workflows), version control and validation (maintaining model versioning and monitoring real-world performance), and appropriate compute infrastructure (cloud versus on-premise depending on volume and budget) [8, 17, 22].

Multidisciplinary team development

Digital twin implementation necessitates collaboration across traditionally siloed disciplines. Core team members include clinical specialists (interventional cardiologists, electrophysiologists, heart failure clinicians, cardiac surgeons), cardiac imaging experts, biomedical engineers and computational scientists, data scientists and AI specialists, clinical informaticists, and regulatory and quality assurance personnel [77–80]. Cross-training initiatives and digital twin heart team conferences embedding technology within clinical decision-making culture are essential [31].

Pilot studies and incremental scaling

Prudent implementation follows stepwise progression: (1) silent-mode retrospective validation, running simulations on historical cases with known outcomes without influencing care, building confidence before patient exposure [8, 22]; (2) prospective observational pilot, presenting twin results as supplementary information, tracking whether recommendations are followed and outcomes; (3) pragmatic clinical trial, randomising cases to twin-guided versus standard care with pre-specified clinical endpoints; (4) operational integration and continuous improvement, embedding twins into routine workflows with ongoing performance monitoring [20, 74].

Limitations of the review

This narrative review, whilst comprehensive in scope, bears inherent limitations that warrant acknowledgement.

As a narrative rather than systematic review, our literature search and selection process lack the exhaustive, protocol-driven rigour of systematic methodologies (e.g. PRISMA guidelines). We did not employ multiple independent reviewers screening all identified records against

Table 3 Pragmatic implementation priorities for cardiac digital twins based on current evidence

Application	Evidence base	Implementation priority	Key considerations
CRT planning [55]	Prospective cohort ($n=45$); 80–85% accuracy; validated clinical endpoint (LVESV reduction)	High	Cardiac MRI + ECG/ECGi pipeline; MDT review; track response rates vs historical controls
CT-FFR [57]	Multiple RCTs ($n>5{,}000$ combined); NICE-approved; cost-effective	Highest (already clinical practice)	Established commercial platforms; reimbursement available; benchmark for digital twin adoption
AF ablation guidance [42, 46]	Feasibility cohorts (< 50 patients); no RCT comparing patient outcomes	Moderate	Embed simulation results in EP lab mapping; focus on persistent AF or redo ablation
HF monitoring [20]	Retrospective validation; no RCT of twin-guided vs standard management	Moderate (research priority)	Wearable data integration; pre-specified alert thresholds; prospective trial essential
Structural heart [17, 66]	Device R&D use established; patient-specific outcome data limited	Lower (complex cases only)	Complex/high-risk TAVR or TMVR cases; centre imaging expertise critical

explicit inclusion/exclusion criteria, register a review protocol a priori, or conduct meta-analyses of quantitative outcomes. Consequently, selection bias towards higher-profile publications, English-language sources, and readily accessible journals may have influenced included evidence.

Formal records of the total number of articles screened, a count of excluded records with reasons, and a structured data extraction table were not maintained prospectively during the review process. This represents a significant methodological limitation: readers cannot verify the comprehensiveness of literature coverage or assess whether included studies are representative of the available evidence. Future reviews in this field should prospectively record screening decisions using PRISMA-compliant documentation and maintain a structured data extraction table capturing key study-level variables (sample size, study design, reference standard, primary outcome, validation approach). The absence of these records limits the reproducibility and transparency of the current synthesis.

The rapidly evolving nature of cardiac digital twin research means evidence published after our search cut-off (September 2025) is not captured. Most included studies represent early-phase translational research, proof-of-concept demonstrations, small cohort validations, and single-centre experiences, with limited prospective randomised controlled trials providing high-level evidence. Publication bias favouring positive feasibility results over negative findings likely inflates perceived performance.

We did not conduct formal quality assessment of included studies using validated tools (e.g. ROBINS-I for observational studies, Cochrane Risk of Bias for trials). Industry funding and conflicts of interest, whilst noted where disclosed, may influence reported outcomes, a concern particularly relevant given commercial digital twin platform involvement in multiple studies.

We did not exhaustively analyse cost-effectiveness, health economic outcomes, or implementation science perspectives. Our review reflects predominantly Western, high-income healthcare contexts; evidence from low- and middle-income countries remains sparse. The scope of clinical applications covered, electrophysiology, heart failure, CRT, coronary assessment, and structural interventions, excludes acute cardiovascular care, primary prevention, and congenital heart disease; this is acknowledged as a deliberate scope limitation explained in the Introduction.

Suggested approaches to overcoming these limitations in future research include (1) conducting PRISMA-compliant systematic reviews with pre-registered protocols and structured data extraction for individual digital twin application domains; (2) designing prospective randomised controlled trials with patient-centred primary endpoints (arrhythmia recurrence, HF hospitalisation, mortality, quality of life) rather than technical model performance metrics; (3) mandating reporting of subgroup performance analyses stratified by sex, ethnicity, and comorbidity burden to address algorithmic bias concerns; (4) establishing international registries prospectively collecting digital twin usage data, predictions, and outcomes in clinical practice; (5) developing consensus reporting standards for digital twin validation studies, analogous to STARD for diagnostic accuracy studies, to enable future meta-analyses; and (6) prioritising studies from diverse healthcare settings, including lower-resource environments, to assess generalisability and equity of access.

Conclusion

Digital twin technologies represent a transformative frontier in cardiovascular medicine, offering a paradigm shift from population-level risk stratification and empirical therapy selection towards precision, patient-specific care. By synthesising multi-modal clinical data with mechanistic biophysical models and artificial intelligence, cardiac digital twins enable in silico experimentation: testing virtual interventions, predicting treatment responses, and personalising therapy decisions before committing to real-world actions.

Current evidence demonstrates feasibility and growing clinical utility across multiple domains, but the level of evidence varies substantially. CT-FFR, the most mature application, has achieved regulatory approval, clinical adoption, and cost-effectiveness demonstration through large multicentre RCTs. CRT digital twin planning has reached moderate translational readiness with prospective cohort evidence demonstrating clinically meaningful outcome prediction. AF and VT ablation guidance, HF monitoring, and structural heart planning remain primarily at proof-of-concept and retrospective validation stages, with patient-centred outcome evidence absent or limited to surrogate endpoints.

A critical message of this review is the distinction between technical modelling performance and patient-centred clinical benefit. Demonstrating that a digital twin can predict an arrhythmia circuit or correlate with a haemodynamic measurement does not establish that acting on twin-guided recommendations improves patient outcomes. Prospective randomised trials comparing digital twin-guided versus standard care, using endpoints of AF-free survival, VT recurrence, HF hospitalisation, symptoms, quality of life, and survival, are urgently needed across all application domains and should be the primary research priority for the field.

Translating research prototypes into routine clinical tools demands addressing substantial challenges: validation extending to longitudinal prediction accuracy and rare event capture; interoperability constraints and workflow integration; ethical imperatives encompassing algorithmic bias mitigation, data privacy governance, and equitable access. Pragmatic implementation roadmaps prioritise high-value, evidence-supported use cases where clinical need is substantial and translational readiness moderate-to-high.

The next translational leap requires collaborative effort across multiple stakeholders: clinicians identifying unmet needs; computational scientists advancing model fidelity and credibility; regulatory agencies refining frameworks; payers establishing reimbursement pathways; industry partners scaling commercial platforms with open standards; and patients engaging in governance ensuring technologies serve their interests equitably. As computational power increases, data ecosystems mature, and evidence accumulates, cardiac digital twins stand poised to transition from promising research tools to integral components of precision cardiovascular medicine, realising the longstanding aspiration of truly personalised, predictive, and preventative care tailored to each patient's unique physiology and disease trajectory.

Abbreviations

CRT	Cardiac resynchronisation therapy
CT	Computed tomography
CT-FFR	Computed tomography-derived fractional flow reserve
ECGi	Electrocardiographic imaging
EMA	European Medicines Agency
FDA	Food and Drug Administration
GPU	Graphics processing unit
MRI	Magnetic resonance imaging
NICE	National Institute for Health and Care Excellence
VT	Ventricular tachycardia
VVUQ	Verification, validation, and uncertainty quantification

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DBO conceptualized the study and reviewed the final manuscript. MD & AO prepared Table 1, DBO & IJO conducted the literature search and prepared Fig. 1. All authors (DBO, MD, AO, MAO, IJO) reviewed and approved the final manuscript.

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